Cheng Cheng

PhD Candidate in Software Engineering, Concordia University

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PROFILE

PhD candidate in Software Engineering at Concordia University. Research focuses on software security, vulnerability management, and the security evaluation of LLM-generated code. Current work studies how LLM-generated software can be benchmarked, detected, repaired, and verified through executable functional and security tests.

EDUCATION

PhD in Software Engineering | Concordia University, Montreal, QC, Canada | 2019 - Present

Supervisor: Prof. Jinqiu Yang.

Research areas: software security, vulnerability management, LLM-generated code, vulnerability detection and repair.

Master of Software Engineering | Concordia University, Montreal, QC, Canada | 2017 - 2019

Bachelor of Software Engineering | Concordia University, Montreal, QC, Canada | 2010 - 2015

RESEARCH INTERESTS

Software security; security of LLM-generated code; vulnerability detection and repair; vulnerability management; secure code generation; empirical software engineering; benchmark construction and evaluation; AI agents; story visualization; object detection; computer vision.

PUBLICATIONS

First-Author Papers and Manuscripts

Detect-Confirm-Repair-Verify for LLM-Generated Software Projects: A Multi-Domain, Multi-Language Empirical Study.
Cheng Cheng.
Status: Revised manuscript submitted to Empirical Software Engineering.

First submission (2026; under an earlier title): arXiv:2603.23633

CFCEval: Evaluating Security Aspects in Code Generated by Large Language Models.
Cheng Cheng, Jinqiu Yang.
In 2025 2nd IEEE/ACM International Conference on AI-powered Software (AIware; co-located with ASE 2025). IEEE, 2025.

Benchmarking the Security Aspect of Large Language Model-Based Code Generation.
Cheng Cheng, Jinqiu Yang.
In LLM4Code 2024: Workshop on Large Language Models for Code (co-located with ICSE 2024), 2024.

A Survey on Software Vulnerability Management.
Cheng Cheng, Jinqiu Yang.
Status: Rejected; currently being rewritten.

Co-Authored Publications

An Exploratory Study on Fine-Tuning Large Language Models for Secure Code Generation.
Junjie Li, Farah Rabbi, **Cheng Cheng**, Aseem Sangalay, Yuan Tian, Jinqiu Yang.
Empirical Software Engineering, 31(4), Article 81, 2026.

Harmonizing Stochasticity and Determinism: Scene-Responsive Diverse Human Motion Prediction.
Tuo Wang, Zhenbo Song, Zhenyu Lou, Qiongjie Cui, Luoming Zhang, **Cheng Cheng**, Haofan Wang, Xu Tang, Huaxia Li, et al.
Advances in Neural Information Processing Systems, NeurIPS 2024.

Fine Tuning Large Language Model for Secure Code Generation.
Junjie Li, Aseem Sangalay, **Cheng Cheng**, Yuan Tian, Jinqiu Yang.
Proceedings of the 2024 IEEE/ACM First International Conference on AI Foundation Models and Software Engineering (FORGE 2024), 2024.

RESEARCH EXPERIENCE

PhD Researcher | Concordia University, Montreal, QC, Canada | *2019 - Present*

- Designed security-oriented benchmarks for evaluating LLM-generated software artifacts.
- Constructed functional and exploit test suites to assess secure-and-correct behavior.
- Studied vulnerability detection, vulnerability repair, and detect-repair-verify workflows for LLM-generated code.
- Conducted empirical studies across multiple programming languages and prompt granularities.
- Prepared literature reviews and empirical studies on software vulnerability management and secure code generation.

TEACHING EXPERIENCE

Teaching Assistant | Concordia University, Montreal, QC, Canada | *2019 - 2020*

- Presented course material and supported students in programming and software engineering topics.
- Graded student projects and assignments.
- Supported course administration and student learning activities.

INDUSTRY EXPERIENCE

Software Testing Intern | ECVictor Inc. | *July 2017 - September 2017*

- Tested APIs and software functionalities.
- Worked with Selenium, Cucumber, JUnit, JMeter, TestNG, Postman, Jenkins, and RestAssured.

IT Manager, Part-time | Leader Aviation | *August 2016 - June 2017*

- Developed and maintained company web pages and websites.
- Supported SEO, software maintenance, system maintenance, and client data support.
- Developed applications using AngularJS and Ionic.

TECHNICAL SKILLS

Programming: JavaScript, Python, PHP, Java, SQL, HTML/CSS
Software Engineering: Web application development, REST APIs, MVC architecture, software testing, empirical software engineering
Security and Evaluation: Vulnerability detection, vulnerability repair, exploit testing, benchmark construction, secure code generation
Tools and Frameworks: Node.js, Express, Flask, SQLite, Git, Docker, Selenium, JUnit, JMeter, Postman, Jenkins
AI and LLMs: LLM-based code generation, LLM-based vulnerability analysis, prompt-based evaluation, empirical evaluation of generated code

LANGUAGES

Chinese Mandarin; English; French.